

Code-Switch ASR Benchmark Report

Project: Voice Triage Hint - agentic voice triage for code-switching patients

Challenge: MLC (Africa) x Intron Agentic Voice AI Challenge, Deep Learning Indaba 2026

Date: 2026-08-06

Clips evaluated: 27

1. Methodology and fairness

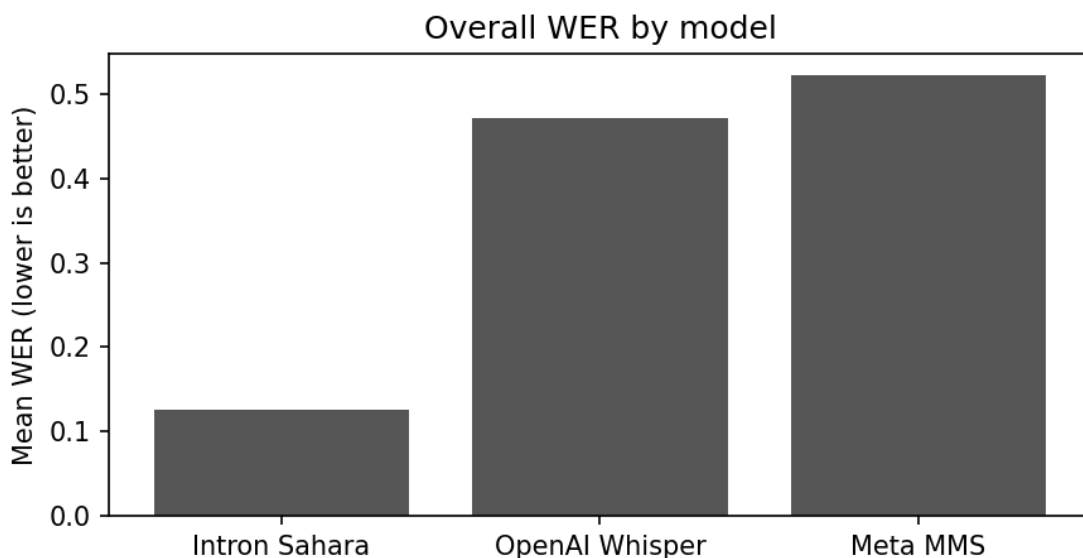
Three speech-to-text models were compared on the same code-switched audio clips: Intron Sahara (commercial API optimized for African accents), OpenAI Whisper (open-source, run locally), and Meta MMS mms-1b-all (open-source, run locally).

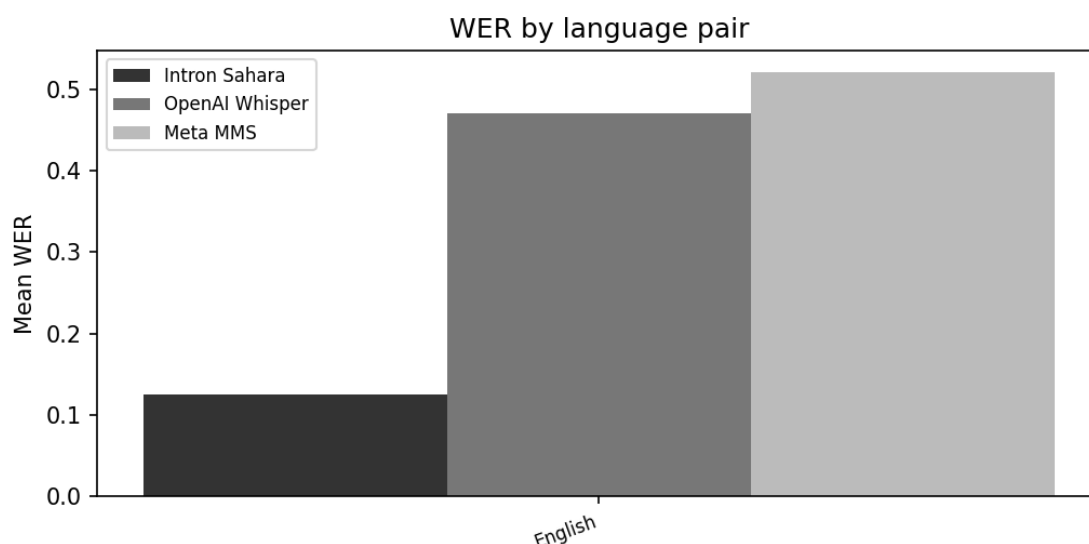
Fairness controls: every model received the identical audio file, converted to 16 kHz mono WAV where required. Every model received the same input-language hint (Sahara: use_language_asr_input; Whisper: language parameter where the language is supported, otherwise auto-detect; MMS: the matching language adapter). Hypotheses and references were normalized identically before scoring: lowercased, punctuation removed, whitespace collapsed (jiwer). Metrics: Word Error Rate (WER), Character Error Rate (CER) - CER is included because it is fairer to agglutinative languages such as Swahili and Kinyarwanda - and wall-clock latency measured around each model call on the same machine (API latency includes network time; local latency depends on hardware).

Test data: self-recorded code-switched clips described in data/samples/metadata.csv with language pair, domain, accent/country, device type, and noise condition per clip.

2. Overall results

Model	Mean WER	Mean CER	Mean latency (s)	Clips
Intron Sahara	0.1255	0.0539	8.3592	26
OpenAI Whisper	0.4712	0.2443	0.8815	27
Meta MMS	0.5219	0.2592	4.0693	27





3. Results by language pair

Language pair: English

Model	Mean WER	Mean CER	Mean latency (s)	Clips
Intron Sahara	0.1255	0.0539	8.3592	26
OpenAI Whisper	0.4712	0.2443	0.8815	27
Meta MMS	0.5219	0.2592	4.0693	27

4. Results by noise condition

Noise condition: corpus recording

Model	Mean WER	Mean CER	Mean latency (s)	Clips
Intron Sahara	0.1255	0.0539	8.3592	26
OpenAI Whisper	0.4712	0.2443	0.8815	27
Meta MMS	0.5219	0.2592	4.0693	27

5. Per-clip transcripts

afrispeech_sw_0.wav | English | swahili (KE) | afrispeech-200 corpus | corpus recording

Reference: Usually, the patient should be identified using two methods.

Intron Sahara (WER 0.2222, CER 0.2069, 8.33s): Usually, the patient should be identified using

OpenAI Whisper (WER 0.3333, CER 0.2586, 0.57s): Usually, called Martha Pissons should be identified using two methods.

Meta MMS (WER 0.2222, CER 0.1207, 7.67s): usually comma the patient should be identified using two method

afrispeech_sw_1.wav | English | swahili (KE) | afrispeech-200 corpus | corpus recording

Reference: Adverse effects First-dose hypotension and postural hypotension are adverse effects.

Intron Sahara (WER 0.0, CER 0.0, 7.07s): Adverse effects: First-dose hypotension and postural hypotension are adverse effects.

OpenAI Whisper (WER 0.5, CER 0.1707, 0.48s): Adverse effects fast hyphen, dose, hypertension and postural hypertension are adverse effects.

Meta MMS (WER 0.6, CER 0.1463, 4.14s): adverse affects first hypen dos hypotentian and postural hypotentian are adverse effects

afrispeech_sw_2.wav | English | swahili (KE) | afrispeech-200 corpus | corpus recording

Reference: A footpath and National Cycle Network cycleway run alongside the Portishead Railway and along the old towpath .

Intron Sahara (WER 0.2353, CER 0.1009, 6.78s): A footpath and National Cycle Network cycleway run along the Potters Head

Railway and also the old towpath.

OpenAI Whisper (WER 0.5294, CER 0.2661, 0.55s): A footpath and national psychonetric cycleway run along the potis head railway and also the old tourpath for stop.

Meta MMS (WER 0.5294, CER 0.2018, 4.46s): a footpath and national cycle network cycle way run along the potys head railway and also the old tor path fullstop

afrispeech_ha_0.wav | English | hausa (NG) | afrispeech-200 corpus | corpus recording

Reference: However, data are not sufficient to draw conclusions about the vitamins effects on human cancers.

Intron Sahara (WER 0.0, CER 0.0, 7.49s): However, data are not sufficient to draw conclusions about the vitamin's effects on human cancers.

OpenAI Whisper (WER 0.2, CER 0.1684, 0.51s): However, coma data are not sufficient to draw conclusions about the vitamins effects on human cancers full stock.

Meta MMS (WER 0.3333, CER 0.1895, 3.6s): however coma data are not sufficient to draw conclusions about the vita means effects on human cancers full stop

afrispeech_ha_1.wav | English | hausa (NG) | afrispeech-200 corpus | corpus recording

Reference: Unspecified superficial injury of unspecified elbow, initial encounter. TABLET, ORAL CIPROFLOXACIN HYDROCHLORIDE, CIPROFLOXACIN HYDROCHLORIDE, EQ 750MG BASE. Pedestrian with other conveyance injured in collision with pedal cycle

Intron Sahara (WER 0.0741, CER 0.0405, 12.62s): Unspecified superficial injury of unspecified elbow, initial encounter. Tablet, oral ciprofloxacin hydrochloride, ciprofloxacin hydrochloride, eq 750mg base. Pedestrian with other conveyance injured in collision with pain.

OpenAI Whisper (WER 0.7407, CER 0.2613, 1.52s): On spacefights, superficial injury of on spacefights elbow, initial encounter, tablet oral cyp pro, plug the synhydrochloride cyp pro, plug the synhydrochloride, 8, 250 mg ways. Pedestrian with other conveyance injured in collision with pedal cyp pro.

Meta MMS (WER 0.5926, CER 0.1396, 7.47s): unspacified superficial injury of unspacified elbow inicial encounter tablet oral cypro pluxasin hydrochlorite cyprofluxasin hydrochlorite each wu s mgbs pedestrian with other convenience injured in colletion with padal cyclo

afrispeech_ha_2.wav | English | hausa (NG) | afrispeech-200 corpus | corpus recording

Reference: PULMONARY WEDGE { a/v/m } 21/22/20

Intron Sahara (WER 0.75, CER 0.2308, 12.48s): Pulmonary wage, um, 21/22/2023.

OpenAI Whisper (WER 1.25, CER 0.6154, 2.84s): ordu with abm 24 22

Meta MMS (WER 1.75, CER 0.8077, 3.16s): a wa aw tt i a ace

afrispeech_yo_0.wav | English | yoruba (NG) | afrispeech-200 corpus | corpus recording

Reference: A second impact tore open the second of the two watertight compartments flooding it.

Intron Sahara (WER 0.0, CER 0.0, 6.59s): A second impact tore open the second of the two watertight compartments, flooding it.

OpenAI Whisper (WER 0.2857, CER 0.1446, 2.81s): A second impact saw open the second of the two watertight compartments, floating first 11

Meta MMS (WER 0.5, CER 0.2289, 3.79s): a second impact tall open the second of te two water tight compartments floding neats foolstop

afrispeech_yo_1.wav | English | yoruba (US) | afrispeech-200 corpus | corpus recording

Reference: X-ray examination of the mastoids is of little value though it may show areas of bone destruction and enlargement of jugular foramen.

Intron Sahara (WER 0.0909, CER 0.0382, 11.15s): X-ray examination of the mastoid is of little value, though it may show areas of bone destruction and enlargement of the jugular foramen.

OpenAI Whisper (WER 0.3636, CER 0.1374, 0.94s): X iPhone Ray examination of the Master 8 is of little value, though it may show areas of bone destruction and enlargement of juggler for army.

Meta MMS (WER 0.3636, CER 0.1985, 3.95s): ex iphonra examination of the mastod is of litl value though it may show areas of bon destruction and enlargement of jugla faramin foodstop

afrispeech_yo_2.wav | English | yoruba (NG) | afrispeech-200 corpus | corpus recording

Reference: Confederate infantry assaults

Intron Sahara (WER 0.0, CER 0.0, 8.5s): Confederate infantry assaults.

OpenAI Whisper (WER 0.0, CER 0.0, 0.4s): confederate infantry assaults.

Meta MMS (WER 0.3333, CER 0.1034, 2.72s): confederate infantry azalts

afrispeech_ig_0.wav | English | igbo (NG) | afriSpeech-200 corpus | corpus recording

Reference: To grant such a patent license to a party means to make such an agreement or commitment not to enforce a patent against the party.

Intron Sahara (WER 0.04, CER 0.0078, 9.82s): To grant such a patent licence to a party means to make such an agreement or commitment not to enforce a patent against the party.

OpenAI Whisper (WER 0.08, CER 0.0775, 0.57s): To grant such a patent license to a party means to make such an agreement or commitment not to enforce a patent against the party full stop.

Meta MMS (WER 0.12, CER 0.093, 5.81s): to grant such a patent license to a party means to make such an agreement or commitment not to enforce a patent against e party full stop

afrispeech_ig_1.wav | English | igbo (NG) | afriSpeech-200 corpus | corpus recording

Reference: He stated, Some were discriminated against while others were completely disenfranchised and their teeming supporters, who were amongst the delegates, could not cast their votes while the exercise lasted.

Intron Sahara (WER 0.0, CER 0.0, 10.91s): He stated, some were discriminated against while others were completely disenfranchised, and their teeming supporters, who were amongst the delegates, could not cast their votes while the exercise lasted.

OpenAI Whisper (WER 0.3793, CER 0.196, 0.67s): stated comma, somewhere discriminated against while others were completely disenfranchised and their teeming supporters comma, where amongst the delegates comma could not cast their votes while the SSI's last two stops.

Meta MMS (WER 0.4138, CER 0.1608, 5.19s): he stated comer some where discriminated against walle others were completely disenfranchised and their teeming supporters comer we were amongst the delegates comer could not cas their votes wile the exercised last foto

afrispeech_ig_2.wav | English | igbo (NG) | afriSpeech-200 corpus | corpus recording

Reference: Tuesday 28 April, 1998 next line. Neuro: Pt Daramola Achike 08 February, 2003

Intron Sahara (WER 0.2308, CER 0.1507, 10.78s): Tuesday, 28 April, 1998

Neuro: Pt Daramola Achike 08 February, 200

OpenAI Whisper (WER 0.5385, CER 0.3562, 0.7s): Tuesday 28 April 1998, Next Line Full Stop. New York colon, Pichandaramola Acheke, 0-8 February 2003.

Meta MMS (WER 1.2308, CER 1.1233, 3.62s): tusday twenty eight april coma nineteen ninety eight next line full stop neuro colon pishen daramola achike zr eig february koma two thusath

afrispeech_zu_0.wav | English | zulu (ZA) | afriSpeech-200 corpus | corpus recording

Reference: Check the product labeling for correct amount of productneeded Figure 1.

Intron Sahara (WER 0.1818, CER 0.0141, 9.56s): Check the product labeling for correct amount of product needed. Figure 1.

OpenAI Whisper (WER 0.5455, CER 0.2394, 0.47s): Check the product labeling for correct amount of product needed to figure one full stop.

Meta MMS (WER 0.6364, CER 0.2394, 3.27s): check the product labelling for correct amount of product needed t figure one full stop

afrispeech_zu_1.wav | English | zulu (LS) | afriSpeech-200 corpus | corpus recording

Reference: The military should not alienate them by further high-handedness.

Intron Sahara (WER 0.0, CER 0.0, 7.58s): The military should not alienate them by further high-handedness.

OpenAI Whisper (WER 0.2222, CER 0.1587, 0.5s): The military should not alienate them by further high-handedness. Full stop.

Meta MMS (WER 0.3333, CER 0.1905, 3.08s): the military should not elienate them by fearther highhandedness foolstop

afrispeech_zu_2.wav | English | zulu (ZA) | afriSpeech-200 corpus | corpus recording

Reference: Dense old scars may undergo hyaline degeneration and subsequent calcification.

Intron Sahara (WER 0.1, CER 0.039, 8.77s): Benz old scars may undergo hyaline degeneration and subsequent calcification.

OpenAI Whisper (WER 0.7, CER 0.2338, 0.47s): Benzodes calls me under boi-line generation and subsequent calcification.

Meta MMS (WER 0.6, CER 0.1299, 3.72s): bensold scars mai undergo highlined generation and subsequent calscification

afriSpeech_af_0.wav | English | Afrikaans (ZA) | afriSpeech-200 corpus | corpus recording

Reference: The forum said it still awaits the actual cause of his death.

Intron Sahara: ERROR - Could not reach the Intron API: HTTPSConnectionPool(host='infer.voice.intron.io', port=443): Max retries exceeded with url: /file/v1/upload/sync (Caused by NameResolutionError("HTTPSConnection(host='infer.voice.intron.io', port=443): Failed to resolve 'infer.voice.intron.io' ([Errno -3] Temporary fa

OpenAI Whisper (WER 0.0833, CER 0.1, 0.53s): The forum said it still awaits the actual cause of his death first.

Meta MMS (WER 0.25, CER 0.15, 3.08s): the forum said it still awaits the actual cause of his death first t

afriSpeech_af_1.wav | English | Afrikaans (ZA) | afriSpeech-200 corpus | corpus recording

Reference: It's completely improbable that each of the 12 women who testified misunderstood their night with Mr. McKnight, Crown prosecutor Katherine Fraser told the jury Monday during closing arguments.

Intron Sahara (WER 0.0357, CER 0.0266, 6.65s): It's completely improbable that each of the 12 women who testified misunderstood their night with Mr. McNair, Crown Prosecutor Katherine Fraser told the jury Monday during closing arguments.

OpenAI Whisper (WER 0.2857, CER 0.1649, 0.62s): It's completely improbable that each of the twelve women testified misunderstood in night with Mr. Midnight Comer. Crown prosecutor Catherine Fraser told during Monday during closing arguments.

Meta MMS (WER 0.3929, CER 0.2128, 4.65s): it's completely improbable that each of the twelve women testified misunderstood their knight with mr mcniel komer crown prosecutor getther and praze the tolto jury manday during closing arguments st

afriSpeech_af_2.wav | English | Afrikaans (ZA) | afriSpeech-200 corpus | corpus recording

Reference: Earlier speaking, on behalf of state Governor Dr Abdullahi Umar Ganduje, the Secretary to the State Government, Alhaji Usman Alhaji, stated the Kano State Government's readiness toward curtailing the menace in the state.

Intron Sahara (WER 0.0909, CER 0.0327, 7.86s): Earlier, speaking on behalf of State Governor Dr. Abdullahi Umar Ganduje, the Secretary to the State Government, Alh. Usman Alh., stated the Kano State Government's readiness towards curtailing the menace in the state.

OpenAI Whisper (WER 0.3636, CER 0.1028, 1.36s): Earlier speaking, and behalf of State Governor Dr. Abdullah Y. Umar Gantujay, the Secretary to the State Government, Ahadi Usman Ahadi, the state of the Khan of State Government's readiness towards curtailing the menace in the state.

Meta MMS (WER 0.4545, CER 0.2009, 6.2s): earlier speaking kome aan behalf of state governor dr abdullahi umargantuj kome i sekretary to the state government komer hahaji usman ahaji komer stated tha canal state government's readiness towards curtailing the menace in the state fulsta

afriSpeech_ig_0.wav | English | Luganda (UG) | afriSpeech-200 corpus | corpus recording

Reference: Since the degree of effect produced by a drug is generally a function of the amount administered, we can express this relationship in terms of a dose-response curve.

Intron Sahara (WER 0.0714, CER 0.0556, 6.74s): Since the degree of effect produced by the drug is generally a function of the amount administered, we can express this relationship in terms of a dose-response.

OpenAI Whisper (WER 0.1429, CER 0.0741, 0.59s): Since the degree of effect produced by the drug is generally a function of the amount administered, we can express this relationship in terms of those responsive...

Meta MMS (WER 0.1429, CER 0.0802, 5.34s): since the degree of effect produced by the drug is generally a function of the amount administered we can express this relationship in terms of those responsive c

afriSpeech_ig_1.wav | English | Luganda (UG) | afriSpeech-200 corpus | corpus recording

Reference: The application of cold therapy reduces bleeding and hematoma formation.

Intron Sahara (WER 0.2222, CER 0.0143, 6.67s): The application of cold therapy reduces bleeding and hematoma formation.

OpenAI Whisper (WER 0.8889, CER 0.3571, 0.48s): The application of code therapy which is just bleeding and hematoma formation will stop.

Meta MMS (WER 0.6667, CER 0.1857, 3.27s): ae application of col therapy reuses bleeding and himatoma formationolsto

afriSpeech_ig_2.wav | English | Luganda (UG) | afriSpeech-200 corpus | corpus recording

Reference: In addition, the capped venous access device will remain patent.

Intron Sahara (WER 0.1111, CER 0.082, 9.99s): In addition, the Captivous Access Device will remain patent.

OpenAI Whisper (WER 0.6667, CER 0.459, 0.48s): In addition, comma, the captivatenius access device remains patent for stove.

Meta MMS (WER 0.6667, CER 0.3934, 3.0s): in addition come to captavarias access device remain patent false

afrispeech_rw_0.wav | English | kinyarwanda (RW) | afrispeech-200 corpus | corpus recording

Reference: Apart from the paucity of funding and insecurity in parts of the country, successive governments fail to perform.

Intron Sahara (WER 0.3333, CER 0.1712, 6.74s): Apart from the bounty of funding and security in parts of the country, successive government variables took performance.

OpenAI Whisper (WER 0.5556, CER 0.4324, 0.51s): Apart from the cost of funding and security, the success of the government varies to the company.

Meta MMS (WER 0.5, CER 0.2793, 3.83s): apart from the post of funding and securituraspasso the tak with success governments fires to perform

afrispeech_rw_1.wav | English | kinyarwanda (RW) | afrispeech-200 corpus | corpus recording

Reference: I'm not the first person to think he should run for higher office, by the way.

Intron Sahara (WER 0.0, CER 0.0, 7.83s): I'm not the first person to think he should run for higher office, by the way.

OpenAI Whisper (WER 0.0625, CER 0.0533, 0.5s): I'm not the first person to think he should learn for higher office by the way.

Meta MMS (WER 0.1875, CER 0.0933, 3.32s): i'm no the first personer to think he should learn for higher office by the way

afrispeech_rw_2.wav | English | kinyarwanda (RW) | afrispeech-200 corpus | corpus recording

Reference: All of them, especially the phalanges, are slightly concave on the inferior plantar side.

Intron Sahara (WER 0.0, CER 0.0, 6.71s): All of them, especially the phalanges, are slightly concave on the inferior plantar side.

OpenAI Whisper (WER 0.7857, CER 0.5, 0.58s): All of them come, especially for land-ease, I slightly conquer one day in the area, plant-as-like period.

Meta MMS (WER 0.5714, CER 0.4186, 3.1s): all of them koman especially for landves kommen are slightly conquerv on the inferior plantaside period

afrispeech_pcm_0.wav | English | pidgin (NG) | afrispeech-200 corpus | corpus recording

Reference: She was selected Best Young Player awards in 2000 season.

Intron Sahara (WER 0.2, CER 0.0893, 6.56s): She was selected Best Young Player Award in the 2000 season.

OpenAI Whisper (WER 0.2, CER 0.1786, 0.48s): She was selected Best Young Player Awards in 2000 season. Full stop.

Meta MMS (WER 0.4, CER 0.3929, 2.55s): she was selected best young player awards in two thousand season foll stop

afrispeech_pcm_1.wav | English | pidgin (NG) | afrispeech-200 corpus | corpus recording

Reference: Management in primary hyperparathyroidism, surgical removal of the parathyroids is curative.

Intron Sahara (WER 0.2727, CER 0.1, 6.61s): Management and primary hyperparathyroidism: surgical removal of the parathyroids is correct.

OpenAI Whisper (WER 1.0909, CER 0.4222, 0.55s): On it meant I'm primary hyperpilot deraising, so the car removal of the pirate words is great.

Meta MMS (WER 0.7273, CER 0.3, 2.63s): management and primary hyperpyadarsin lorgical remover of the paratorixs iscre

afrispeech_pcm_2.wav | English | pidgin (NG) | afrispeech-200 corpus | corpus recording

Reference: Minor anaphylactic reactions such as urticaria can be treated with oral or intramuscular chlorpheniramine.

Intron Sahara (WER 0.0, CER 0.0, 6.55s): Minor anaphylactic reactions such as urticaria can be treated with oral or intramuscular chlorpheniramine.

OpenAI Whisper (WER 0.9286, CER 0.4667, 3.12s): Minor and a Philactic Russian such as the Tishk Korea can be treated with ORAO in drugs and narcissism there.

Meta MMS (WER 0.5714, CER 0.219, 3.25s): minor anafhelatic rasions such as betish cariar can be treated with ora or intramuscular profi ni raman

6. Limitations and bias notes

The sample set is small and self-recorded by the team, so results indicate trends rather than statistically significant differences. Accent coverage is limited to the speakers available to the team; models may perform differently on other regional accents, ages, and genders. Local model latency is hardware-dependent and not directly comparable to API latency, which includes network round-trips. Reference transcripts were written by the speakers themselves;

transcription conventions for code-switched text (spelling of borrowed words) can bias WER slightly for all models equally. Raw per-clip outputs are provided in reports/results.json for independent verification.